

Luiz Carlos Costa da Silva

AI Backend Engineer

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Portuguese (Native) • English (Advanced) • Mandarin (Learning)

SUMMARY

AI Backend Engineer focused on production AI systems and Python backends. At Auria, I build Python AI agents in production handling 3,000+ daily B2B security incidents, own the data layer across 36 PostgreSQL tables, and built the company's Metabase BI and an on-call alerting bot from scratch. Previously led a multi-year Ruby-on-Rails-to-Kotlin migration at IME-USP's innovation hub (1,000+ patents, 100+ innovation initiatives). B.Sc. in Computer Science from the University of São Paulo (IME-USP); capstone on Retrieval-Augmented Generation (RAG) using LangChain and GPT-4o.

SKILLS

Languages & Frameworks: Python, Kotlin, TypeScript, JavaScript, SQL, FastAPI, Vue.js, React, asyncio, Pydantic

Databases & Infrastructure: PostgreSQL, SQLAlchemy, Redis, Docker, Git, Metabase, Twilio API, Vector Databases

AI / LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Embeddings, LangChain, Pydantic-AI, GPT-4o, Claude

Concepts: Software Architecture, System Design, Microservices, REST APIs, WebSockets, Observability, Unit Testing, CI/CD, OOP, Algorithms & Data Structures

EXPERIENCE

Backend Software Engineer

August 2025 – Present

Auria AI

São Paulo, Brazil

- Developed Python AI agents running in production across the incident-response pipeline, handling **3,000+ daily B2B security incidents** – summarization, insight extraction, and automated decision-making over a fleet of **20+ production agents**.
- Bootstrapped the PostgreSQL/SQLAlchemy data layer, **owning the schema across 36 production tables** and shipping backend features end-to-end across multiple microservices.
- Built **Auria's Metabase BI from scratch** – designed all dashboards, KPI analyses, and threshold-based alerts that incident-response leadership uses daily.
- Designed and shipped an **on-call alerting bot** that detects production drops and **places voice calls (Twilio) to the responsible engineers** with AI-generated incident summaries – cutting overnight outage detection from **8+ hours** (caught the next morning) to near-immediate.
- Tech stack:** Python, FastAPI, asyncio, PostgreSQL, SQLAlchemy, Redis, Pydantic-AI, Metabase, Twilio, Docker.

Software Developer (Research Scholarship)

January 2021 – August 2023

Hub USP Inovação – IME-USP

São Paulo, Brazil

- Migrated a large legacy Ruby on Rails system to a modern, scalable Kotlin backend, redesigning domain models, REST APIs, and automated test suites, and managing deployment infrastructure with Docker.
- Maintained AUSPIN's innovation platform – a Latin American reference for university innovation transparency – serving **1,000+ patents, 3,000+ companies, 3,000+ professors, and 100+ innovation initiatives**.
- Tech stack:** Kotlin, Vue.js, JavaScript, Docker. Advised by Prof. Dr. Alfredo Goldman.

PROJECTS

Divination – RAG Assistant for D&D Dungeon Masters

2024

B.Sc. Capstone Project, IME-USP

- Built a Retrieval-Augmented Generation (RAG) chat assistant grounded in the official 2024 D&D Free Rules, helping Dungeon Masters retrieve and reason over rule content during sessions.
- Designed the retrieval pipeline using LangChain and a vector store for semantic search over the rulebook, integrating GPT-4o for grounded answer generation.
- Implemented a FastAPI backend exposing REST endpoints and a Vue 3 frontend for the chat interface.
- Advised by M.Sc. Renato Cordeiro Ferreira. **Tech stack:** Python, FastAPI, Vue 3, LangChain, GPT-4o, Vector Store.

EDUCATION

B.Sc. in Computer Science

2020 – 2024

University of São Paulo (IME-USP)

LEADERSHIP & ACTIVITIES

President & Member, USPCoDeLab

January 2021 – March 2024

IME-USP

São Paulo, Brazil

- Served as President (Jan 2022 – Aug 2023), leading a team of **20+ members** focused on web-development education and student-led tech initiatives.
- Organized two university hackathons – secured sponsorships, mentored participants, and managed budgets.